DSAI 490 – Assignment 1
Representation Learning with AE & VAE on Medical MNIST

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1 Architectures

Both the AE and VAE share a convolutional backbone on 64×64 grayscale inputs: three strided 3×3 conv blocks ($32 \to 64 \to 128$ channels) followed by a dense projection to a $D = 16$ latent vector, and a mirrored transposed-conv decoder. The **AE** produces a deterministic code $z = E(x)$ and minimizes mean-squared error $\mathcal{L}_{\mathrm{AE}} = \mathbb{E}\,\|x - D(z)\|^2$. The **VAE** head instead outputs $(\mu, \log \sigma^2)$; a sample is drawn via the reparameterization trick $z = \mu + \sigma \odot \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$. Its objective combines a per-pixel binary cross-entropy and the Gaussian KL to the prior:

$$\mathcal{L}_{\mathrm{VAE}} = \underbrace{\mathbb{E}_{q(z|x)}[-\log p(x|z)]}_{\text{recon}} + \beta \underbrace{\mathrm{KL}(q(z|x) \,\|\, \mathcal{N}(0, I))}_{\text{KL}}.$$

We linearly anneal β from 0 to 1 over the first five epochs to avoid posterior collapse.

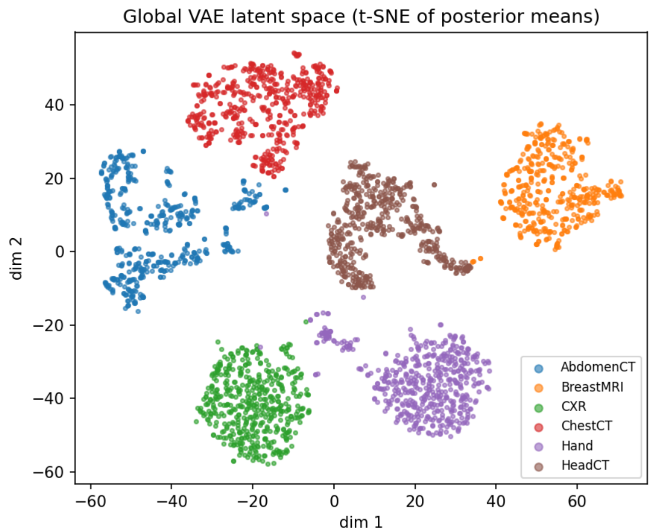

Figure 1: Global VAE latent space, t-SNE projection of posterior means on held-out data. Six anatomical regions form distinct but not isolated clusters, reflecting the smooth $\mathcal{N}(0, I)$ prior that is absent from the deterministic AE.

2 Setup

Medical MNIST (58k images, 6 classes) is streamed via `tf.data` with a fixed 80/10/10 train/val/test split per class. For each of the six anatomical regions we train one AE and one VAE (12 models); we additionally train one *global* AE and one *global* VAE on the union of all classes, used solely for the cross-class latent visualization – per-class encoders have unaligned latent axes, so stacking their codes would produce a misleading plot. All models use Adam ($\mathrm{lr} = 10^{-3}$), batch size 128, and 20 epochs. See `notebook.ipynb` for the full pipeline.

Class	AE MSE	AE SSIM	VAE MSE	VAE SSIM
AbdomenCT	—	—	—	—
BreastMRI	—	—	—	—
CXR	—	—	—	—
ChestCT	—	—	—	—
Hand	—	—	—	—
HeadCT	—	—	—	—

Table 1: Test-set reconstruction metrics (fill from `figures/metrics_per_class.csv`).

3 Latent-space behavior

The AE's latent space is compact but irregular: clusters are present but their relative positions reflect arbitrary encoder geometry. The VAE's prior pulls codes toward $\mathcal{N}(0, I)$, yielding a more continuous manifold where interpolating between two codes decodes to a smooth morph between regions (see the latent-traversal figure in `figures/`).

4 Results

As expected the AE produces sharper reconstructions (lower MSE, higher SSIM) because it optimizes pixel fidelity directly, while the VAE trades a small amount of sharpness for a structured latent that supports *generation* from the prior (see `figures/vae_samples_global.png`) and interpolation. For *denoising*, both models improve on the noisy input at $\sigma \in \{0.1, 0.2, 0.3\}$; the VAE degrades more gracefully at high σ because its prior-regularized decoder resists over-fitting to noise patterns.

5 Insights

KL warm-up was essential: without it, one or two classes drove the VAE into posterior collapse within the first epoch. Per-region reconstruction difficulty ranks, roughly, Hand < CXR < HeadCT < ChestCT < AbdomenCT < BreastMRI, tracking intra-class variability. The cross-class t-SNE (Fig. 1) confirms that the global VAE's latent captures anatomical-region identity without any class supervision – unsupervised representation learning is working as intended.